SkyRun Inc. Customer Retention Analysis

Cohort analysis and customer segmentation, December 2024 to November 2025

Executive summary

SkyRun is worried that too many new customers buy once and never come back. We looked at a year of transaction data for 4,372 customers to find out when customers stop buying, whether this is getting worse, and which customers matter most.

The short answer is that SkyRun loses most of its customers straight after their first purchase. Three out of four new customers do not buy anything in the following month, and about half have not come back three months later. Customers who do make a second purchase tend to stick around, so the real job is getting people from their first order to their second.

A steady decline in retention over the year was not found as return rates stayed low but stable, although customers who joined between February and May 2025 were worthless in their first six months than those who joined in January. Revenue depends heavily on a small group of loyal buyers, that is 18% of customers bring in 57% of revenue, while the largest group, 35% of customers, has already stopped buying.

Headline findings

- **Most customers are lost after the first order:** On average, only 25% of a new cohort buys again in the next month.
- **About half don't come back within three months:** The average three-month return rate was 46%.
- **Retention is low, not falling:** Month-two retention stayed between 19% and 30% across cohorts, with no clear downward trend.
- **There is a clear window to act:** Returning customers came back after a median of 41 days, and 71% of them returned within 90 days.
- **Value is concentrated:** Champions (18% of customers) generate 57% of revenue. Lost customers are the biggest group at 35%.
- **A third of customers are drifting away:** At the end of the period, 13.7% were at risk and 19.5% had churned.

Top recommendations

1. **Build a second-order programme:** Follow up with every new customer in their first 30 to 45 days, using personalised recommendations and a reason to come back.
2. **Step in before 90 days:** Send re-engagement offers to customers who have not bought for 90 days, starting with the 597 currently at risk.
3. **Look after the Champions:** They carry more than half the revenue, so losing even a few of them is costly. A loyalty or VIP scheme is worth testing.

Background and data

The business problem

SkyRun is an online retailer selling fashion, electronics, home goods, beauty and wellness products. It has been good at attracting new customers, but many of them make one purchase and do not return. This lowers customer lifetime value and makes the money spent on acquiring customers harder to justify. SkyRun wanted to understand when customers drop off and where it should focus its retention efforts.

The data

The dataset contains 541,909 transaction lines from 1 December 2024 to 9 December 2025. Each line records the invoice number, date, customer ID, product code, product category, quantity, unit price and country.

About a quarter of the lines (135,080, or 24.9%) had no customer ID. These were removed because they cannot be linked to a customer or placed in a cohort. That left 406,829 lines covering 4,372 customers, 22,190 orders and 9,000 products. We found no duplicate rows, returns or zero-price items.

Data quality issues

Some parts of the data needed careful handling

- **Country cannot be used.** Country is attached to each transaction rather than each customer, and the typical customer appears in 22 of the 28 countries. The countries also do not match SkyRun's US and UK markets, so we did not compare regions.
- **Product category cannot be used either.** The data has ten categories, such as Books, Toys and Electronics, but each has almost the same number of sales lines, and the typical product appears under all ten. Because the same product is recorded under different categories, category does not reliably show what a customer bought, so retention by first-purchase category was not analysed.
- **The last month is incomplete.** December 2025 only has 9 days of data, so it was left out of the retention analysis.
- **The first cohort is not really new.** Anyone who was already a customer before December 2024 shows up in the December 2024 cohort, which makes that cohort look better than the others. We left it out when comparing cohorts.

Methodology

The analysis was done in Python using pandas, matplotlib, seaborn and scikit-learn. The main steps were

1. **Cohorts.** Each customer was assigned to the month of their first purchase. We then tracked how many customers from each cohort bought again in each later month.
2. **Fair comparisons.** Newer cohorts have had less time to come back, so we always compared cohorts over the same period: retention at months 2, 4 and 7, the share returning within three months, and revenue per customer in the first six months.

3. **Drop-off and timing.** We measured how long repeat customers took to place a second order. Second orders placed on the same day as the first (10.9% of cases) were excluded, as they are more likely to be split baskets than genuine return visits.
4. **Churn definition.** Since 90% of returning customers came back within 182 days, customers inactive for longer than that were classed as churned, and those inactive for 90 to 182 days as at risk.
5. **Segmentation.** We calculated recency, frequency and monetary value (RFM) for each customer. Because frequency and spend were heavily skewed, the values were log-transformed and standardised before K-Means clustering. Four clusters were chosen: the silhouette scores for three and four clusters were almost identical (0.310 and 0.307), and four gave more useful groups.

Key findings

1. Most customers are lost straight after their first purchase

The heatmap shows, for each cohort, the share of customers who bought again in each month after joining. The biggest drop happens immediately. On average, only 25% of a cohort buys again in the month after joining, a fall of 75% points. After that, retention barely changes, averaging between 24% and 28% each month up to month eight.

This tells us that once someone becomes a repeat buyer, they usually keep buying. The problem is getting them there in the first place.

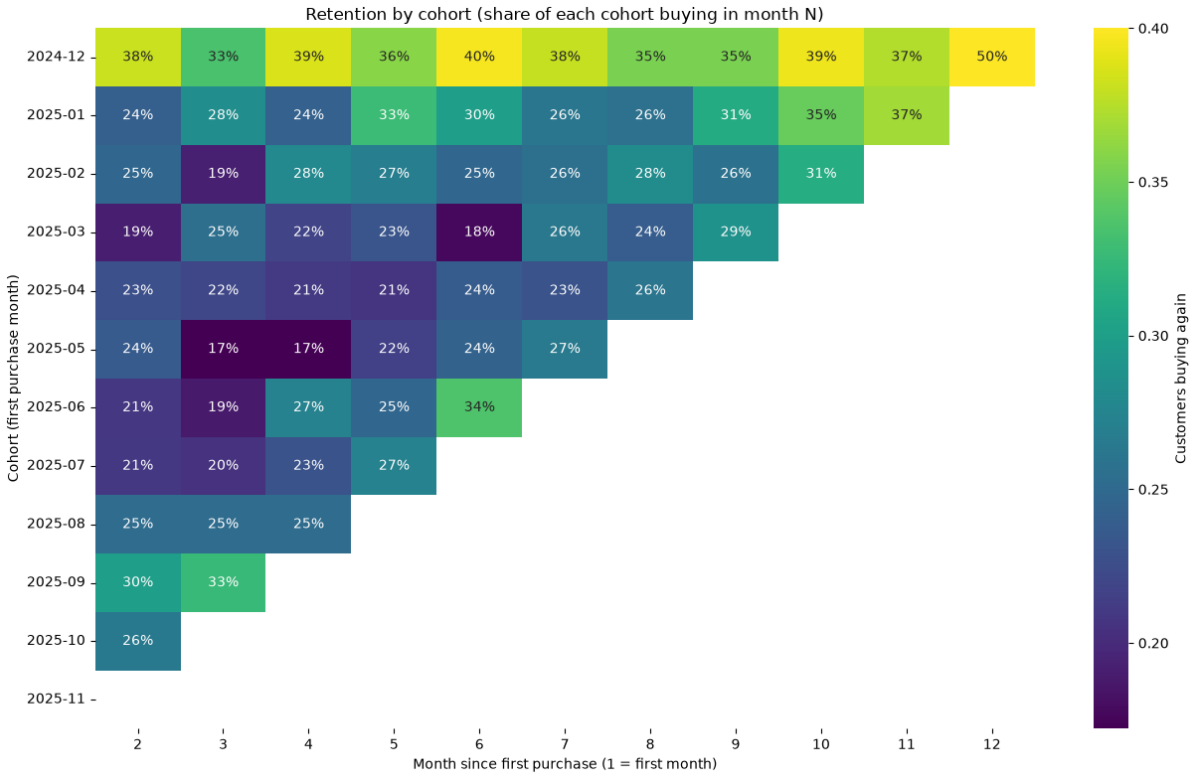


Figure 1. Retention by cohort, complete months only. The December 2024 cohort is higher because it includes existing customers, and the last cell in each row falls in November, the seasonal peak.

2. Retention is low, but it is not getting worse

We expected to see newer cohorts retaining fewer customers, but the data does not show that. Month-two retention for the January to October 2025 cohorts averaged 23.7% and moved between 19% and 30% without a clear trend. Retention at months four and seven told the same story.

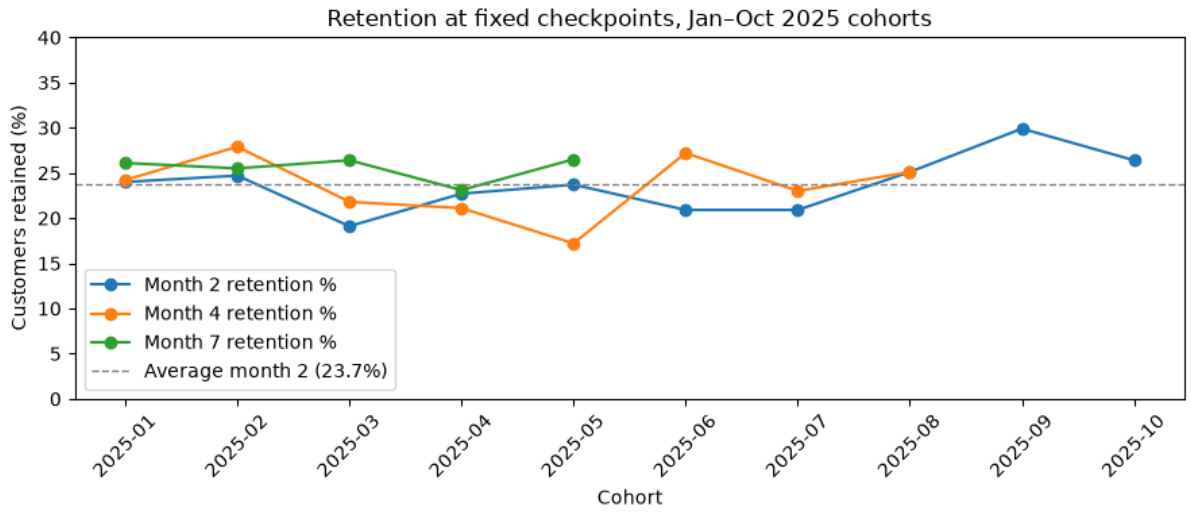


Figure 2. Retention at fixed checkpoints by cohort. The December 2024 cohort is excluded.

Looking at whether customers came back at any point in their first three months gives a similar picture. Around half returned, with the March to July cohorts doing slightly worse:

Cohort	Returned within 3 months
January 2025	50.6%
February 2025	49.5%
March 2025	45.0%
April 2025	46.5%
May 2025	39.8%
June 2025	45.1%
July 2025	44.5%
August 2025	49.7%

Over the whole year, 70% of customers did buy again at some point. So, most customers are not lost forever, just come back rarely and irregularly, which is why monthly retention looks so low.

3. Customers who come back usually do so within three months

Among customers who placed a second order, the median wait was 41 days. Around 42% came back within 30 days and 71% within 90 days. After about three months, very few return on their own. This gives SkyRun a clear window for follow-up.

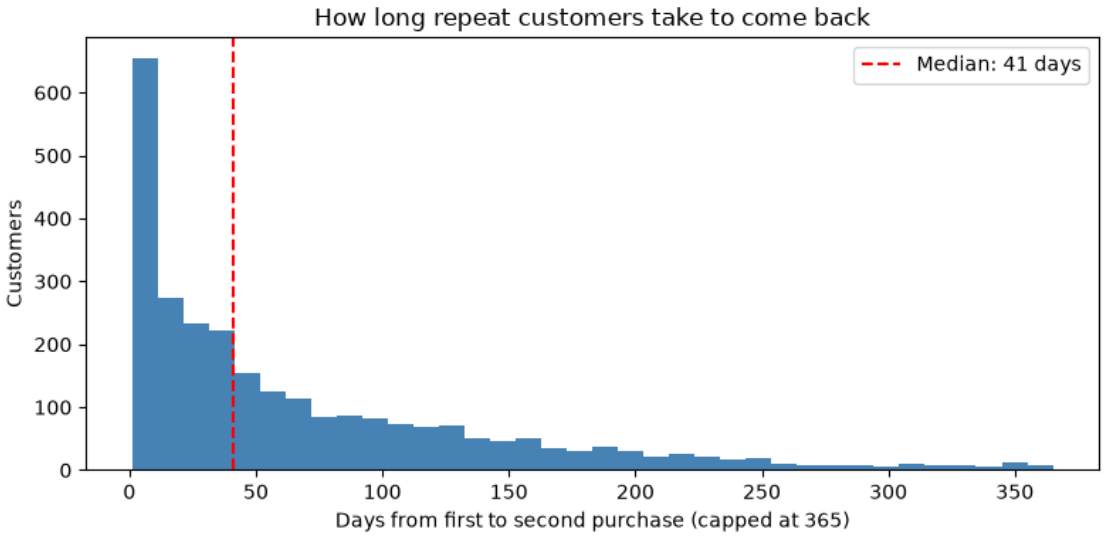


Figure 3. Days between first and second purchase for repeat customers. Second orders placed on the same day as the first are excluded.

4. A third of the customer base is lapsing or gone

Using the 90-day and 182-day thresholds, this is how customers stood at the end of the period:

Status	Customers	Share
Active (bought in the last 90 days)	2,922	66.8%
At risk (90 to 182 days)	597	13.7%
Churned (more than 182 days)	853	19.5%

The active figure is flattered a little by the autumn sales peak, when many new customers joined. Some of them have only bought once so far.

5. A small group of customers drives most of the revenue

The clustering produced four clear segments:

Segment	Days since last purchase	Orders	% of customers	% of revenue
Champions	11	15.7	18.0%	56.9%
Regulars	74	4.8	26.4%	27.9%
New / Promising	21	2.3	20.8%	8.5%
Lost	190	1.5	34.8%	6.8%

- **Champions** buy often, recently and bring in more than half of all revenue.
- **Regulars** buy steadily but have not ordered for 74 days on average, so many are close to the at-risk point.
- **New / Promising** customers bought recently but only two or three times. They are the main group to convert into regular buyers.

- **Lost** customers have not bought for over six months. They are the biggest group but contribute very little revenue.

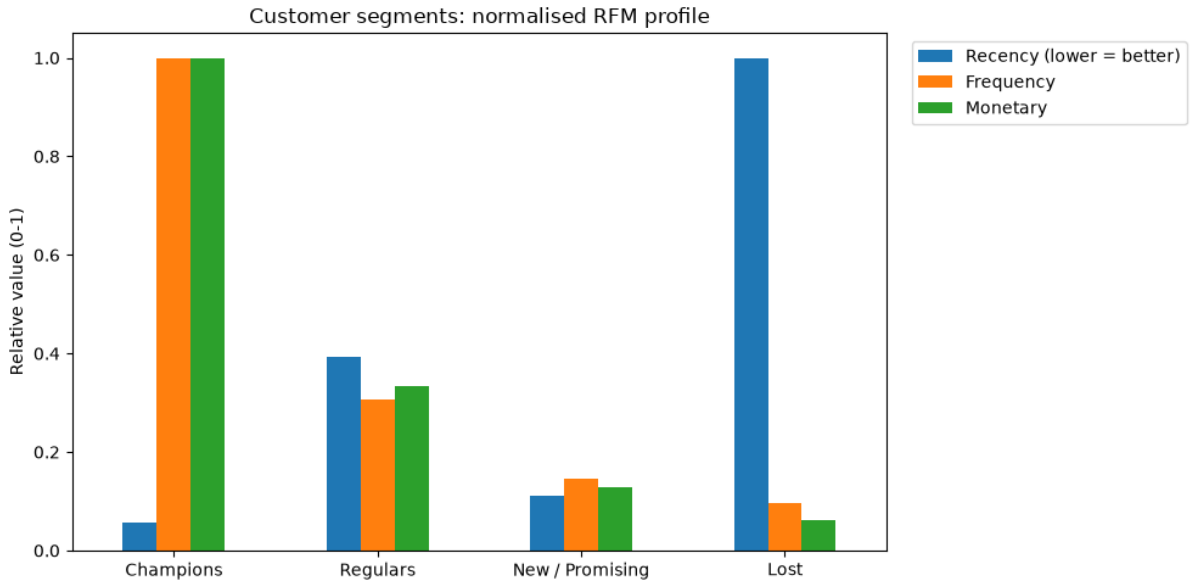


Figure 4. RFM profile of each segment, scaled from 0 to 1. For recency, a shorter bar is better.

Frequency and spend were strongly linked (correlation of 0.78), so the customers who order most often are also the ones who spend the most.

6. Customers who joined between February and May 2025 were worth less in their first six months

Comparing revenue per customer over each cohort's first six months, customers who joined between February and May 2025 spent 20% to 26% less than January's customers. The May cohort was the weakest, which matches its low three-month return rate. The June cohort recovered, but its six months include the November peak, so part of that is seasonal.

Cohort	Customers	6-month revenue per customer	Change vs January
January 2025	421	158,588	–
February 2025	380	127,273	-20%
March 2025	440	127,617	-20%
April 2025	299	126,328	-20%
May 2025	279	117,624	-26%
June 2025	235	143,094	-10%

This is the closest thing in the data to the decline SkyRun described. It is a modest drop over a short period, and it is mixed up with seasonal effects, so it should be watched rather than treated as a confirmed trend.

Recommendations

1. **Focus on the second order:** This is where most customers are lost. Set up an automatic series of messages for new customers. Example; a thank-you and recommendations in the first week, a personalised offer around day 30, and a final nudge before day 45. Track the three-month return rate for each new cohort, starting from today's average of about 46%.
2. **Re-engage customers at 90 days:** A customer who has not bought for 90 days is moving into the at-risk group. Start with the 597 at-risk customers and the Regulars who are approaching this point. These customers already know SkyRun, so they should be cheaper to bring back than finding new ones.
3. **Protect the Champions:** With 57% of revenue coming from 18% of customers, SkyRun should make sure this group feels valued, for example through loyalty rewards, early access to sales or priority service. Watch for any Champion who goes quiet for longer than usual.
4. **Keep win-back for Lost customers cheap:** Lost customers are the largest group but add little revenue. Low-cost email campaigns are worth trying, but heavy discounts for this group are unlikely to pay off.
5. **Find out why February to May 2025 customers were weaker:** Check which campaigns and channels brought in the February to May cohorts. If certain promotions attract one-off bargain hunters, budget could move towards channels that bring in customers who return.

Limitations and next steps

Limitations

- **No usable location or category data.** Country and product category are recorded inconsistently for the same customer or product, so retention could not be compared by region or by first purchase category.
- **Only one year of data.** SkyRun described a decline over 18 months, but the data covers about 12, so the longer trend could not be tested. With one year, it is also hard to separate cohort differences from seasonal effects.
- **Missing customers.** A quarter of transactions had no customer ID. If these anonymous shoppers behave differently, the true retention rate could be lower than shown here.
- **No product or marketing data.** There is no record of how customers were acquired, so retention could not be compared by marketing channel. Product category could not be used either, for the reason given under data quality.
- **Churn is estimated.** Customers do not tell SkyRun when they leave, so churn is based on inactivity. Because the data only covers a year, long gaps between purchases are under-represented, and the 182-day threshold may be on the low side.

Next steps

1. Repeat the analysis with reliable customer locations and at least 18 months of data.
2. Add product category and acquisition channel, so retention can be compared by what customers first bought and where they came from.

3. Test the second-order programme properly, with a control group that does not receive the messages, so its effect can be measured.
4. Once more data is available, build a churn prediction model to flag customers at risk before they reach 90 days.